

ARA-Net: Atlas-Guided Multimodal Alzheimer's Disease Staging with Locked External Subject-Level Validation and Structural Neurodegeneration Consistency

Yuanqin Zhao^{*a,b,c}, Yu Zhang^d, Jinghong Hu^a, Jian Mao^e, Bisong Zhou^a,
Zhuoyao Song^a, Tengfei Guo^{†b}, and Shaohua Ma^{†a,c,f}

^aInstitute of Biopharmaceutical and Health Engineering (iBHE), Tsinghua Shenzhen International Graduate School (SIGS), Tsinghua University, Shenzhen 518055, China.

^bInstitute of Neurological and Psychiatric Disorders, Shenzhen Bay Laboratory, Shenzhen 518132, China.

^cKey Laboratory of Industrial Biocatalysis, Ministry of Education, Tsinghua University, Beijing 100084, China.

^dInstitute of Integrated Circuits, School of Integrated Circuits, Tsinghua University, Beijing 100084, China.

^eCentre de Recherche sur l'Inflammation, INSERM-U1149, Université Paris-Cité, Faculté de Médecine Xavier Bichat, Paris 75018, France.

^fKey Laboratory of Active Proteins and Peptides Green Biomanufacturing of Guangdong Higher Education Institutes, Tsinghua Shenzhen International Graduate School, Shenzhen 518055, China.

Abstract

Background. Structural MRI and clinical variables provide complementary information for Alzheimer's disease (AD) staging, but externally useful models require leakage-

^{*}First author. Email: zhao_yuanqin@163.com

[†]Corresponding authors.

aware subject-level evaluation, explicit uncertainty analysis, and biologically bounded interpretation. We developed ARA-Net V6 as a multi-cohort, atlas-guided multimodal framework for CN/MCI/AD staging with locked external heldout testing and atlas-based structural neurodegeneration consistency analysis.

Methods. We constructed subject-level splits across ADNI, AIBL, IXI, and OASIS. ADNI was used for training, validation, and internal testing; AIBL was divided into adaptation training, adaptation validation, and a locked heldout external test; IXI served as a healthy negative-control cohort; and OASIS was retained only as an external stress test. The final algorithm is RC-SPE, a risk-constrained subject-level probability ensemble. RC-SPE combines six base-model probability streams using log-probability pooling, non-negative model weights, class-specific offsets, temperature scaling, and subject-level probability averaging. Biological evidence was evaluated as atlas structural neurodegeneration consistency in a priori AD-relevant regions.

Results. Final subject-level RC-SPE achieved AIBL heldout accuracy of 0.903, BAcc of 0.833, macro AUC of 0.937, AD-vs-CN AUC of 1.000, and CN/MCI/AD recall of 0.961/0.686/0.852. Bootstrap 95% confidence intervals were 0.759–0.899 for BAcc, 0.894–0.974 for macro AUC, 0.531–0.839 for MCI recall, and 0.710–0.966 for AD recall. IXI healthy CN retention was 1.000. AIBL errors were concentrated near MCI/AD boundaries, with no AD endpoint unit misclassified as CN. OASIS transfer remained weak and is reported as a limitation. The AIBL heldout AD-key atlas-volume consistency score was 0.510 versus a uniform regional null of 0.286, with bootstrap CI 0.479–0.526 and permutation $p = 0.026$.

Conclusion. The revised work supports a domain-adapted, subject-level, atlas-guided multimodal AD staging framework with strong locked AIBL heldout performance, preserved IXI healthy specificity, explicit residual MCI uncertainty, and atlas-region structural neurodegeneration consistency. It does not claim pure zero-shot transfer, solved OASIS generalization, direct Braak-stage validation, or deployment-ready clinical performance.

Keywords: Alzheimer’s disease; structural MRI; atlas-guided multimodal learning; external validation; subject-level staging; neurodegeneration consistency; open-source research prototype

Abbreviations: AD, Alzheimer’s disease; ADNI, Alzheimer’s Disease Neuroimaging Initiative; AIBL, Australian Imaging, Biomarkers and Lifestyle study; AUC, area under the receiver-operating-characteristic curve; BAcc, balanced accuracy; CAS, Clinical Alignment Score; CN, cognitively normal; IXI, Information eXtraction from Images dataset; MCI, mild cognitive impairment; MMSE, Mini-Mental State Examination; OASIS, Open Access Series of Imaging Studies; sMRI, structural magnetic resonance imaging.

1 Introduction

Structural MRI is widely used in AD research because it captures neurodegeneration patterns including medial temporal atrophy, ventricular enlargement, and broader brain-volume change (Jack et al., 2018; Frisoni et al., 2010). These anatomical changes are not sufficient for clinical diagnosis by themselves, but they provide a biologically meaningful substrate for machine-learning models that aim to support CN/MCI/AD staging. A credible staging model must therefore demonstrate not only within-cohort accuracy, but also leakage-aware external evaluation, class-specific error behavior, and biological interpretation that is proportional to the evidence actually available.

We designed ARA-Net V6 as an atlas-guided multimodal AD staging framework with explicit cohort roles and subject-level endpoints. ADNI provides development and internal testing data, AIBL provides external-domain adaptation and locked heldout testing, IXI provides a healthy negative-control specificity endpoint, and OASIS is retained as a stress-test cohort. This design separates model fitting, external evaluation, negative-control specificity, and residual transfer testing.

The final model target is locked AIBL heldout subject-level CN/MCI/AD classification with bootstrap uncertainty and transparent MCI/AD error analysis. The algorithmic core is RC-SPE, a risk-constrained subject-level probability ensemble that combines heterogeneous base-model probability streams while balancing external BAcc, MCI/AD recall, calibration, AD-to-CN error avoidance, and IXI healthy specificity.

The biological analysis is intentionally bounded. Because direct neuropathological labels are not available, the manuscript tests whether atlas structural MRI changes concentrate in a priori AD-relevant regions. This analysis is a structural neurodegeneration consistency check, not a proof of postmortem disease stage or attention-map biomarker discovery.

The revised contributions are:

- C1.** A leakage-aware subject-level protocol across ADNI, AIBL, IXI, and OASIS.
- C2.** Direct external CN/MCI/AD classification on a locked AIBL heldout endpoint.
- C3.** IXI healthy negative-control evaluation to quantify false impairment predictions.
- C4.** RC-SPE, a risk-constrained subject-level probability ensemble with non-negative model weights, class offsets, temperature scaling, and subject-level probability averaging.
- C5.** Bootstrap stability analysis and subject-level MCI/AD error analysis.
- C6.** Replacement of the unsupported attention-only CAS/Braak claim with atlas structural neurodegeneration consistency analysis.
- C7.** An open-source research deployment package with explicit clinical-use boundaries.

Figure 1 summarizes the study architecture as a multi-cohort, subject-level AD staging framework: data roles are explicit, the atlas-guided multimodal input is separated from the RC-SPE algorithm, and each validation endpoint has a defined interpretation boundary.

2 Methods

2.1 Study design and notation

The revised study uses a multi-cohort, subject-level design. Each scan-level record is represented as

$$r_i = (d_i, s_i, t_i, y_i, x_i), \quad (1)$$

where d_i is the dataset, s_i is the subject identifier, t_i is the visit or scan identifier, $y_i \in \{\text{CN}, \text{MCI}, \text{AD}\}$ is the diagnostic label when available, and x_i is the multimodal feature vector. The full data inventory is

$$\mathcal{D} = \mathcal{D}_{\text{ADNI}} \cup \mathcal{D}_{\text{AIBL}} \cup \mathcal{D}_{\text{IXI}} \cup \mathcal{D}_{\text{OASIS}}. \quad (2)$$

The development and evaluation roles were fixed before final endpoint reporting:

$$\mathcal{D}_{\text{ADNI}} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{val}} \cup \mathcal{D}_{\text{internal}}, \quad \mathcal{D}_{\text{AIBL}} = \mathcal{D}_{\text{adapt-train}} \cup \mathcal{D}_{\text{adapt-val}} \cup \mathcal{D}_{\text{heldout}}. \quad (3)$$

IXI was used as a healthy negative-control cohort, and OASIS was held out as an external stress test. OASIS was not used for final model tuning.

Subject-level leakage control was imposed by requiring disjoint subject identifiers between development and evaluation partitions:

$$\mathcal{S}_a \cap \mathcal{S}_b = \emptyset \quad \forall a \neq b, \quad a, b \in \{\text{train}, \text{val}, \text{internal}, \text{adapt-train}, \text{adapt-val}, \text{heldout}\}, \quad (4)$$

where $\mathcal{S}_a = \{s_i : r_i \in \mathcal{D}_a\}$. This rule is important because repeated imaging from the same participant can otherwise inflate performance estimates (Varoquaux et al., 2017).

2.2 Cohort construction and endpoint inventory

Table 1 summarizes the split inventory. The subject counts in the table are unique split-level participants used to define leakage-free splits. For longitudinal cohorts, endpoint units can differ from unique participants because a participant may contribute repeated scans or distinct diagnostic-state endpoint units over time. The primary endpoint is therefore described explicitly as subject-level diagnostic-state classification after probability aggregation.

Placeholder for Figure 1. Study design and data architecture.

Panels to draw: **(a)** multi-cohort study map with ADNI, AIBL, IXI, and OASIS roles; **(b)** scan and subject inventory across all cohorts; **(c)** ADNI and AIBL split strategy; **(d)** subject-level leakage control and scan-to-endpoint aggregation; **(e)** endpoint hierarchy; **(f)** overall analysis flow from data to RC-SPE prediction and bounded biological interpretation.

Figure 1: Study design and data architecture. **(a)** The study uses four cohorts with non-overlapping roles: ADNI for model development and internal testing, AIBL for external-domain adaptation and locked heldout testing, IXI for healthy negative-control specificity, and OASIS for external stress testing. **(b)** The full split inventory contains 4,388 scans and 2,023 split-level subjects, including ADNI train/validation/internal-test sets, AIBL adaptation and heldout sets, 581 IXI healthy subjects, and 99 OASIS stress-test subjects. **(c)** ADNI is split into training, validation, and internal testing; AIBL is split into adaptation training, adaptation validation, and locked heldout testing. **(d)** Subject-level leakage control keeps subject identifiers disjoint across development and evaluation partitions, and repeated scans are aggregated into subject-level diagnostic-state endpoints. **(e)** The primary endpoint is AIBL locked heldout subject-level CN/MCI/AD staging; IXI evaluates healthy-control CN retention; OASIS is interpreted only as a stress-test endpoint. **(f)** The analysis flow links cohort construction, atlas-guided multimodal features, RC-SPE probability integration, subject-level prediction, error analysis, and atlas-based structural neurodegeneration consistency.

Table 1: V6 split inventory. Counts refer to scan inventory and unique split-level subjects.

Split	Scans	Subjects	Role
ADNI train	1,686	450	Main training
ADNI validation	355	97	Validation and calibration
ADNI internal test	360	96	Internal risk analysis
AIBL adaptation training	719	385	External-domain adaptation
AIBL adaptation validation	191	105	External-domain calibration
AIBL locked heldout	397	210	Primary external test
OASIS external	99	99	Stress test
IXI healthy external	581	581	Healthy negative-control specificity

For a subject-level diagnostic-state unit u , scan indices were grouped as

$$\mathcal{I}_u = \{i : s_i = s(u), y_i = y(u), d_i = d(u), r_i \in \mathcal{D}_{q(u)}\}, \quad (5)$$

where $q(u)$ is the split assignment. This grouping prevents scan-level repetitions from being interpreted as independent final endpoint units. Scan-level results are retained only as reference analyses.

2.3 Atlas-guided multimodal feature representation

MRI features were extracted from a 21-region atlas and included regional volumetric and intensity summaries. The atlas representation is denoted

$$a_i = [a_{i,1}, a_{i,2}, \dots, a_{i,p_a}] \in \mathbb{R}^{p_a}. \quad (6)$$

The core clinical feature vector is denoted

$$c_i = [\text{age}_i, \text{sex}_i, \text{education}_i, \text{APOE4}_i, \text{MMSE}_i, \text{CDR-SB}_i] \in \mathbb{R}^{p_c}, \quad (7)$$

where unavailable variables were imputed within the training/adaptation workflow and accompanied by missingness indicators when required. The final multimodal representation was

$$x_i = [\tilde{a}_i, \tilde{c}_i, m_i] \in \mathbb{R}^p, \quad (8)$$

where $\tilde{a}_i$ and $\tilde{c}_i$ are standardized atlas and clinical features, and m_i contains missingness indicators.

Feature standardization used development-set statistics only:

$$\tilde{x}_{i,j} = \frac{x_{i,j} - \mu_j^{(\text{dev})}}{\sigma_j^{(\text{dev})} + \eta}, \quad (9)$$

where η is a small numerical constant. The development statistics were not recomputed on locked heldout endpoint labels.

The model is therefore atlas-guided and multimodal. The atlas component anchors the model to interpretable neuroanatomical regions, while the clinical component reflects the known diagnostic signal contained in age, global cognition, and functional staging. Figure 2 shows the feature system. The central evidence is classification performance and structural consistency, not attention-map biomarker discovery.

2.4 Candidate models and comparator roles

The revised experimental framework included atlas-only, cascade, atlas+clinical, clinical-only, biomarker-enhanced, and ensemble models. These models have different interpretive roles. Atlas-only and cascade models test whether MRI-derived regional structure alone can provide sufficient external staging. Atlas+clinical models test the main scientific target: anatomically grounded multimodal staging. Clinical-only models act as upper-bound comparators because clinical variables can encode strong diagnostic signal but do not retain the MRI atlas-guided mechanism. Biomarker-enhanced variants provide sensitivity analyses when additional variables are available.

For each base model m , class probabilities were represented as

$$p_m(y = k | x_i) = \frac{\exp(f_{m,k}(x_i))}{\sum_{c \in \mathcal{Y}} \exp(f_{m,c}(x_i))}, \quad \mathcal{Y} = \{\text{CN}, \text{MCI}, \text{AD}\}. \quad (10)$$

When a model family produced calibrated probabilities directly, $f_{m,k}$ denotes its logit-scale representation. When a tree-based model produced probability estimates, the probabilities were clipped before log-pooling to avoid numerical underflow.

2.5 RC-SPE: risk-constrained subject-level probability ensemble

The final locked algorithm was RC-SPE, a risk-constrained subject-level probability ensemble. RC-SPE was designed to avoid a common failure mode in external AD staging: improving aggregate accuracy while sacrificing minority disease recall or overcalling impairment in healthy external controls. Figure 3 provides the visual architecture of the algorithm. RC-SPE combined six base-model probability streams:

- AIBL-adapted atlas-biomarker-enhanced HGB.
- AIBL-adapted atlas-core-clinical HGB.
- AIBL-adapted clinical-biomarker-only RF balanced.
- AIBL-adapted clinical-core-only HGB.
- AIBL-adapted clinical-core-only RF balanced.

Placeholder for Figure 2. Atlas-guided multimodal feature system.

Panels to draw: **(a)** T1 MRI preprocessing to atlas-derived regional features; **(b)** 21-region atlas schematic with hippocampus, amygdala, and lateral ventricles highlighted; **(c)** core clinical feature block; **(d)** feature standardization and missingness indicators; **(e)** multimodal feature matrix; **(f)** anatomical link between AD-relevant regions and structural neurodegeneration.

Figure 2: Atlas-guided multimodal feature system. **(a)** Structural T1 MRI is converted into atlas-level regional measurements, including volumetric and intensity summaries from the 21-region representation used by the model. **(b)** The atlas representation includes the full region set and highlights the a priori AD-relevant regions used for structural consistency analysis: bilateral hippocampus, bilateral amygdala, and bilateral lateral ventricles. **(c)** The core clinical block contains age, sex, education, APOE4, MMSE, and CDR-SB when available, reflecting demographic, genetic-risk, cognitive, and functional information. **(d)** Continuous features are standardized using development-set statistics, and missing variables are handled with imputation plus missingness indicators rather than by deleting external subjects. **(e)** The final input matrix concatenates standardized atlas features, standardized clinical features, and missingness indicators for each scan-level record before probability modeling. **(f)** The feature design links the classifier to established AD structural patterns, especially medial temporal atrophy and lateral ventricular enlargement, while keeping the biological claim at the level of atlas structural neurodegeneration consistency.

• RF–logistic regression ensemble component.

Let $p_{m,k}(x_i)$ denote the class probability for scan i , class k , and base model m . The final ensemble uses log-probability pooling:

$$z_{i,k} = \frac{1}{T} \sum_{m=1}^M w_m \log(\max(p_{m,k}(x_i), \epsilon)) + b_k, \quad k \in \{\text{CN}, \text{MCI}, \text{AD}\}, \quad (11)$$

where

$$w_m \geq 0, \quad \sum_{m=1}^M w_m = 1, \quad T > 0. \quad (12)$$

Here w_m are non-negative model weights, T is a temperature parameter, b_k is a class-specific offset, and ϵ prevents taking the logarithm of zero. Calibrated class probabilities are:

$$\tilde{p}_{i,k} = \frac{\exp(z_{i,k})}{\sum_{c \in \mathcal{Y}} \exp(z_{i,c})}. \quad (13)$$

RC-SPE was selected using ADNI validation, AIBL adaptation validation, and IXI healthy-control specificity, with OASIS excluded from the final tuning objective. In compact form, the selection objective can be written as

$$\begin{aligned} \max_{\theta} \quad \mathcal{J}(\theta) = & \alpha_1 \text{BAcc}_{\text{ADNI val}}(\theta) + \alpha_2 \text{AUC}_{\text{ADNI val}}(\theta) + \alpha_3 \text{BAcc}_{\text{AIBL adapt val}}(\theta) \\ & + \alpha_4 \min\{\text{Rec}_{\text{MCI}}^{\text{AIBL adapt val}}(\theta), \text{Rec}_{\text{AD}}^{\text{AIBL adapt val}}(\theta)\} + \alpha_5 \text{Ret}_{\text{CN}}^{\text{IXI}}(\theta), \end{aligned} \quad (14)$$

where $\theta = (w, b, T)$ and the α terms define the balanced rescue profile. This objective reflects the practical aim of rescuing minority-class performance while preserving healthy-control specificity.

Because IXI contains healthy subjects only, we also tracked false impairment as a hard risk indicator:

$$\text{Risk}_{\text{IXI}}(\theta) = 1 - \text{Ret}_{\text{CN}}^{\text{IXI}}(\theta), \quad (15)$$

and treated candidate profiles with improved AIBL MCI recall but increased IXI false impairment as riskier alternatives rather than as the locked final model.

Algorithm 1. RC-SPE for locked subject-level AD staging.

1. Train or load M base probability models and obtain $p_{m,k}(x_i)$ for each scan.
2. Select non-negative weights w_m , offsets b_k , and temperature T on ADNI validation, AIBL adaptation validation, and IXI healthy specificity; do not use OASIS for tuning.
3. Pool probabilities by Eq. 11 and transform logits by Eq. 13.
4. Aggregate repeated scans into subject-level endpoint probabilities by Eq. 16.
5. Lock $\theta = (w, b, T)$ and report AIBL heldout, IXI healthy specificity, OASIS stress-test, bootstrap, calibration, and error analyses.

2.6 Subject-level probability aggregation

For each subject-level endpoint unit u , repeated scan probabilities were averaged:

$$\bar{p}_{u,k} = \frac{1}{|\mathcal{J}_u|} \sum_{i \in \mathcal{J}_u} \tilde{p}_{i,k}, \quad \hat{y}_u = \arg \max_{k \in \mathcal{Y}} \bar{p}_{u,k}. \quad (16)$$

Prediction confidence and decision margin were defined as

$$\text{conf}(u) = \max_k \bar{p}_{u,k}, \quad \text{margin}(u) = \max_k \bar{p}_{u,k} - \max_{c \neq \hat{y}_u} \bar{p}_{u,c}. \quad (17)$$

These quantities were used for error analysis but not to change locked predictions.

2.7 Classification metrics

Let C_{ab} be the confusion-matrix entry for true class a and predicted class b . Per-class recall, precision, and balanced accuracy were defined as

$$\text{Rec}_a = \frac{C_{aa}}{\sum_b C_{ab}}, \quad \text{Prec}_a = \frac{C_{aa}}{\sum_b C_{ba}}, \quad \text{BAcc} = \frac{1}{|\mathcal{Y}_{\text{present}}|} \sum_{a \in \mathcal{Y}_{\text{present}}} \text{Rec}_a. \quad (18)$$

Overall accuracy was

$$\text{Acc} = \frac{\sum_a C_{aa}}{\sum_a \sum_b C_{ab}}. \quad (19)$$

For multiclass AUC, one-vs-rest AUCs were averaged across evaluable classes:

$$\text{MacroAUC} = \frac{1}{|\mathcal{Y}_{\text{AUC}}|} \sum_{k \in \mathcal{Y}_{\text{AUC}}} \text{AUC}(\mathbf{1}[y = k], \bar{p}_k). \quad (20)$$

For IXI, all subjects are healthy controls, so the primary specificity metric was CN retention:

$$\text{Ret}_{\text{CN}}^{\text{IXI}} = \frac{\#\{u \in \text{IXI} : \hat{y}_u = \text{CN}\}}{\#\{u \in \text{IXI}\}}, \quad \text{FIR}^{\text{IXI}} = 1 - \text{Ret}_{\text{CN}}^{\text{IXI}}, \quad (21)$$

Placeholder for Figure 3. RC-SPE locked subject-level probability ensemble.

Panels to draw: **(a)** six base probability streams; **(b)** CN/MCI/AD probability simplex or probability vectors from each stream; **(c)** log-probability pooling with non-negative weights; **(d)** class offsets and temperature scaling; **(e)** risk-constrained objective; **(f)** subject-level probability averaging; **(g)** final CN/MCI/AD output with confidence and margin.

Figure 3: RC-SPE locked subject-level probability ensemble. **(a)** RC-SPE receives six CN/MCI/AD probability streams: AIBL-adapted atlas-biomarker HGB, atlas-core-clinical HGB, clinical-biomarker RF, clinical-core HGB, clinical-core RF, and an RF–logistic regression cascade component. **(b)** Each stream contributes a three-class probability vector rather than a binary decision, allowing uncertainty between CN, MCI, and AD to be preserved before pooling. **(c)** The ensemble performs weighted log-probability pooling with non-negative weights constrained to sum to one, which discourages a single model from dominating the locked decision. **(d)** Class-specific offsets and temperature scaling calibrate the pooled logits and adjust CN/MCI/AD decision balance. **(e)** Model selection balances AIBL adaptation-validation BAcc, minority disease recall, AD-to-CN error avoidance, calibration, and IXI healthy-control CN retention, with OASIS excluded from tuning. **(f)** Repeated scans are averaged into subject-level diagnostic-state probabilities before the primary endpoint is reported. **(g)** The final output is a three-class prediction with calibrated probabilities, maximum-probability confidence, and decision margin for downstream error analysis.

where FIR denotes the false impairment rate.

2.8 Bootstrap uncertainty

Uncertainty for the locked AIBL heldout subject-level endpoint was estimated using 2,000 bootstrap resamples. For a metric M and bootstrap sample b , the empirical estimate was

$$\hat{M}^{(b)} = M\left(\{(y_u, \bar{p}_u) : u \in \mathcal{B}^{(b)}\}\right), \quad \mathcal{B}^{(b)} \sim \text{SampleWithReplacement}(\mathcal{U}_{\text{AIBL heldout}}). \quad (22)$$

The percentile confidence interval was

$$\widehat{\text{CI}}_{95\%}(M) = \left[Q_{0.025}\{\hat{M}^{(b)}\}_{b=1}^B, Q_{0.975}\{\hat{M}^{(b)}\}_{b=1}^B\right], \quad B = 2,000. \quad (23)$$

2.9 MCI/AD error analysis

Subject-level errors were summarized using transition rates:

$$R_{ab} = \frac{C_{ab}}{\sum_c C_{ac}}, \quad a, b \in \{\text{CN}, \text{MCI}, \text{AD}\}. \quad (24)$$

We also defined error groups for adjacent and severe mistakes:

$$\mathcal{E}_{a \rightarrow b} = \{u : y_u = a, \hat{y}_u = b, a \neq b\}. \quad (25)$$

Group profiles were summarized for age, MMSE, CDR-SB, APOE4, atlas hippocampal volume, atlas lateral ventricular volume, AD-like atlas z-score, maximum probability, and decision margin. The purpose was to distinguish boundary-like uncertainty from clinically severe collapse of disease cases into CN.

2.10 Structural neurodegeneration consistency

The biological analysis tests whether disease-associated atlas volume differences concentrate in a priori AD-relevant regions: bilateral hippocampus, bilateral amygdala, and bilateral lateral ventricles. Let $\mathcal{R}$ be the atlas-region set and $\mathcal{K} \subset \mathcal{R}$ be the six AD-key regions. For each region r , the absolute disease-associated volume contrast was

$$\Delta_r = |\mu_{r,\text{AD}} - \mu_{r,\text{CN}}|, \quad (26)$$

where the means were computed within the relevant evaluation group. The AD-key structural consistency score was

$$\text{NCS}_{\mathcal{K}} = \frac{\sum_{r \in \mathcal{K}} \Delta_r}{\sum_{r \in \mathcal{R}} \Delta_r}. \quad (27)$$

The uniform regional null was

$$\text{NCS}_0 = \frac{|\mathcal{K}|}{|\mathcal{R}|} = \frac{6}{21} = 0.286. \quad (28)$$

Permutation testing evaluated whether the observed concentration exceeded random region-set selection:

$$p_{\text{perm}} = \frac{1 + \sum_{b=1}^B \mathbf{1}[\text{NCS}_{\mathcal{K}^{(b)}} \geq \text{NCS}_{\mathcal{K}}]}{B + 1}, \quad |\mathcal{K}^{(b)}| = |\mathcal{K}|. \quad (29)$$

This is a structural MRI neurodegeneration proxy. It does not provide postmortem-stage validation or attention-map biomarker evidence.

2.11 Open-source research deployment and clinical-use boundary

The public implementation includes reproducible analysis scripts, a research inference wrapper, an HTTP API/static frontend, aggregate reports, figures, documentation, and toy probability examples. Raw ADNI, AIBL, OASIS, and IXI data are governed by their original access agreements and are not redistributed. Row-level private predictions, clinical spreadsheets, MRI volumes, and non-redistributable checkpoints are excluded from the public release.

The software is released as an open-source research prototype for retrospective evaluation and future prospective validation. It is not a medical device, is not cleared or approved for clinical use, and is not intended for standalone diagnosis or patient-care decisions.

3 Results

3.1 Comparator baselines and external endpoint overview

The Results are organized around three questions: whether RC-SPE improves locked AIBL subject-level CN/MCI/AD staging relative to comparator baselines, whether healthy external controls are preserved as CN in IXI, and whether OASIS exposes residual transfer limitations. The comparator set includes atlas-only, cascade, ADNI-only hybrid, atlas+clinical, biomarker-enhanced, clinical-only, and RC-SPE variants so that the final model can be interpreted relative to both anatomically constrained and clinically strong baselines.

3.2 Locked AIBL heldout subject-level performance

Final locked RC-SPE achieved strong AIBL heldout subject-level performance, summarized visually in Figure 4. It achieved accuracy 0.903, BAcc 0.833, macro AUC 0.937, AD-vs-CN AUC 1.000, and CN/MCI/AD recall 0.961/0.686/0.852. The scan-level reference result

Table 2: AIBL external model comparison and final RC-SPE positioning. The locked primary endpoint is final RC-SPE at subject level on AIBL heldout.

Model/protocol	Unit	Test cohort	Endpoint n	Acc	BAcc	Macro AUC	AD-vs-CN AUC/CN retention	CN/MCI/AD recall
Old v3 ensemble	scan	AIBL external	1,307	0.606	0.399	0.597	NA	NA
Atlas-only HGB	scan	AIBL heldout	397	0.776	0.479	0.732	AD-vs-CN AUC 0.884	0.944/0.151/0.341
Cascade RF-logreg	scan	AIBL heldout	397	0.751	0.391	0.756	AD-vs-CN AUC 0.886	0.954/0.000/0.220
ADNI-only hybrid RF	scan	AIBL heldout	397	0.549	0.406	0.753	AD-vs-CN AUC 0.870	0.614/0.604/0.000
v4 atlas+clinical HGB	scan	AIBL heldout	397	0.882	0.741	0.942	AD-vs-CN AUC 0.990	0.964/0.528/0.732
Biomarker-enhanced HGB	scan	AIBL heldout	397	0.861	0.703	0.942	AD-vs-CN AUC 0.990	0.957/0.396/0.756
Clinical-only RF comparator	scan	AIBL heldout	397	0.922	0.835	0.957	AD-vs-CN AUC 0.997	0.970/0.755/0.780
Final RC-SPE	scan	AIBL heldout	397	0.909	0.820	0.939	AD-vs-CN AUC 0.998	0.964/0.642/0.854
Final RC-SPE	subject	AIBL heldout	216	0.903	0.833	0.937	AD-vs-CN AUC 1.000	0.961/0.686/0.852
Final RC-SPE	subject	IXI healthy	581	1.000	1.000	NA	CN retention 1.000	1.000/0.000/0.000
Final RC-SPE	subject	OASIS stress	99	0.586	0.334	0.554	AD-vs-CN AUC 0.371	0.966/0.034/0.000

was similar: accuracy 0.909, BAcc 0.820, macro AUC 0.939, AD-vs-CN AUC 0.998, and CN/MCI/AD recall 0.964/0.642/0.854.

IXI healthy negative-control specificity was preserved. The final model retained all IXI healthy subjects as CN at both scan and subject levels, corresponding to CN retention 1.000 and false impairment rate 0.000. This result does not provide AD/MCI staging evidence because IXI contains healthy controls only, but it directly addresses whether the final model overcalls impairment in a healthy external cohort.

3.3 Comparator interpretation

Table 2 shows that the final subject-level RC-SPE is substantially stronger than the atlas-only MRI baseline, cascade baseline, ADNI-only hybrid model, biomarker-enhanced sensitivity model, and earlier atlas+clinical model on the locked AIBL endpoint. The clinical-only RF comparator achieved AIBL heldout BAcc 0.835 and CN/MCI/AD recall 0.970/0.755/0.780 at scan level. This model is reported as a comparator and upper bound rather than the central ARA-Net model, because it does not retain the atlas-guided MRI component central to the revised scientific objective. This comparison is important: it shows that clinical variables carry substantial diagnostic signal and prevents overstating MRI-only interpretability.

RC-SPE is therefore not presented as the single numerically best classifier under every possible feature setting. It is presented as the final model that balances the paper’s core requirements: atlas-guided multimodal staging, high locked AIBL subject-level performance, MCI/AD rescue, and IXI healthy specificity. This distinction should remain explicit in the manuscript and figures.

3.4 Algorithmic ablation and calibration of RC-SPE

We evaluated RC-SPE against single-model, simple averaging, equal log-pooling, partial-parameter, and aggregation ablations using the same private prediction streams. Only aggregate metrics are reported. Figure 6 is designed to visualize these algorithmic comparisons. The ablation demonstrates that the final locked method is not equivalent to a generic average ensemble.

Placeholder for Figure 4. External classification performance.

Panels to draw: **(a)** AIBL BAcc model comparison; **(b)** AIBL macro AUC model comparison; **(c)** Final RC-SPE CN/MCI/AD recall; **(d)** AD-vs-CN discrimination; **(e)** IXI healthy specificity; **(f)** OASIS stress-test performance; **(g)** scan-level versus subject-level RC-SPE comparison.

Figure 4: **External subject-level classification performance.** **(a)** AIBL heldout BAcc compares atlas-only HGB (0.479), cascade RF-logreg (0.391), ADNI-only hybrid RF (0.406), v4 atlas+clinical HGB (0.741), biomarker-enhanced HGB (0.703), clinical-only RF comparator (0.835), final RC-SPE scan-level (0.820), and final RC-SPE subject-level (0.833). **(b)** AIBL macro AUC uses the same model order, showing final RC-SPE subject-level macro AUC 0.937 and clinical-only comparator macro AUC 0.957. **(c)** Per-class recall for final subject-level RC-SPE is CN 0.961, MCI 0.686, and AD 0.852, making MCI the main residual disease-boundary weakness rather than a collapsed class. **(d)** AD-vs-CN discrimination reaches AUC 1.000 at the AIBL subject-level endpoint. **(e)** IXI healthy negative-control analysis shows CN retention 1.000 and false impairment rate 0.000 across 581 healthy subjects. **(f)** OASIS is presented as a compact stress-test panel with BAcc 0.334, macro AUC 0.554, and CN/MCI/AD recall 0.966/0.034/0.000, explicitly separated from the primary external claim. **(g)** Scan-level and subject-level RC-SPE results are shown side by side to emphasize that subject-level averaging is the locked primary endpoint.

Table 3: Algorithmic ablation of RC-SPE. OASIS is reported only as a stress-test metric and was not used for final model selection.

Variant	AIBL BAcc	AIBL AUC	MCI recall	AD recall	AD→CN	IXI CN retention	ECE	OASIS BAcc
Best single base model	0.756	0.945	0.571	0.741	2	0.997	0.122	0.310
Arithmetic mean ensemble	0.711	0.951	0.400	0.741	3	1.000	0.112	0.333
Equal log-pooling	0.648	0.954	0.171	0.778	5	1.000	0.094	0.333
Final weights only	0.815	0.955	0.657	0.815	0	1.000	0.136	0.333
Weights + offsets	0.823	0.926	0.657	0.852	0	1.000	0.219	0.310
Weights + temperature	0.815	0.954	0.657	0.815	0	1.000	0.052	0.333
Full RC-SPE, scan-level	0.810	0.940	0.660	0.805	0	1.000	0.081	0.334
Full RC-SPE, subject-level	0.833	0.937	0.686	0.852	0	1.000	0.078	0.334
MCI-rescue profile	0.811	0.952	0.886	0.593	0	0.959	0.156	0.367
AD-rescue profile	0.698	0.930	0.143	0.963	0	1.000	0.293	0.328

The best single base model achieved AIBL BAcc 0.756, MCI recall 0.571, AD recall 0.741, two AD-to-CN errors, and IXI CN retention 0.997. Equal log-pooling without learned weights, offsets, or temperature achieved AIBL BAcc 0.648 and MCI recall 0.171. By contrast, full RC-SPE at subject level achieved AIBL BAcc 0.833, MCI recall 0.686, AD recall 0.852, zero AD-to-CN errors, and IXI CN retention 1.000.

The calibration analysis further supports the temperature-scaled component of RC-SPE. The best single base model had AIBL expected calibration error (ECE) 0.122 and negative log-likelihood (NLL) 0.416. The final RC-SPE reduced ECE to 0.078 and NLL to 0.320 while improving BAcc. The weights+temperature ablation achieved the lowest ECE (0.052) and NLL (0.298), but its AIBL BAcc remained 0.815 and AD recall 0.815. The locked final RC-SPE therefore prioritizes the combined endpoint of external staging, AD/MCI rescue, zero AD-to-CN errors, and IXI specificity rather than calibration alone.

We also evaluated risk-profile alternatives. A high-MCI-recall profile increased AIBL MCI recall to 0.886 but reduced AD recall to 0.593 and IXI CN retention to 0.959, meaning 24 IXI healthy subjects were predicted as impaired. A high-AD-recall profile increased AD recall to 0.963 but reduced MCI recall to 0.143. These tradeoffs support the risk-constrained final selection. Leave-one-model-out sensitivity remained stable: dropping any one of the six base streams produced AIBL BAcc 0.823–0.835 and preserved zero AD-to-CN errors, indicating that RC-SPE is not dependent on a single fragile base model.

3.5 Subject-level MCI and AD error analysis

On the locked AIBL heldout subject-level set, errors were concentrated at disease-stage boundaries, as planned for Figure 5. Among 154 CN endpoint units, 148 were classified as CN, five as MCI, and one as AD. Among 35 MCI endpoint units, 24 were classified as MCI, two as CN, and nine as AD. Among 27 AD endpoint units, 23 were classified as AD and four as MCI; no AD endpoint unit was misclassified as CN.

The transition rates reinforce the same interpretation. MCI-to-AD errors accounted for 0.257 of MCI endpoint units, whereas MCI-to-CN errors accounted for 0.057. AD-to-MCI errors accounted for 0.148 of AD endpoint units, and AD-to-CN errors were 0.000. This pattern

Table 4: AIBL heldout subject-level confusion matrix for final RC-SPE.

True label	Predicted CN	Predicted MCI	Predicted AD	Recall
CN	148	5	1	0.961
MCI	2	24	9	0.686
AD	0	4	23	0.852

is more clinically plausible than a classifier that preserves global accuracy by assigning most uncertain disease cases to CN.

The feature-profile analysis supported a boundary-error interpretation. AIBL AD endpoint units correctly classified as AD had lower MMSE, larger lateral ventricular volume, and higher AD-like atlas z-scores than AD endpoint units classified as CN/MCI. MCI endpoint units classified as AD had lower MMSE and more AD-like atlas profiles than MCI endpoint units classified correctly, consistent with disease-severity boundary uncertainty rather than arbitrary failure.

Internal subject-level performance remained weaker. Internal subject-level BAcc was 0.448 with CN/MCI/AD recall 0.241/0.553/0.550. This internal calibration tension should not be hidden; it is reported as a limitation and motivates future site-specific calibration work. The key external claim remains the locked AIBL heldout endpoint with IXI specificity.

3.6 Algorithmic stability, risk tradeoff, and bootstrap uncertainty

Bootstrap analysis with 2,000 resamples quantified uncertainty for the locked AIBL heldout subject-level endpoint and is included in Figure 6. The 95% CI for BAcc was 0.759–0.899, and the 95% CI for macro AUC was 0.894–0.974. The 95% CI for MCI recall was 0.531–0.839, and the 95% CI for AD recall was 0.710–0.966. These intervals support the final model as the locked primary result while preserving MCI staging uncertainty as a visible limitation.

The bootstrap result also clarifies which claim is strongest. CN recall and AD-vs-CN separation are robust on AIBL heldout, while MCI recall remains the widest uncertainty interval. MCI staging is improved and externally meaningful, but it remains the main residual uncertainty.

3.7 Structural neurodegeneration consistency and OASIS stress test

The AIBL heldout atlas-volume consistency score exceeded the uniform regional null in the a priori AD-key regions, as summarized in Figure 7. The score was 0.510 versus a uniform null of 0.286, with score difference 0.225, bootstrap CI 0.479–0.526, and permutation $p = 0.026$.

Across all labeled AD-relevant data, the AD-key consistency score was 0.426 versus a uniform null of 0.286, with permutation $p = 0.0207$. The ADNI-only internal check remained non-significant, with score 0.342 and $p = 0.1843$. These results support a bounded structural MRI neurodegeneration proxy while making clear that postmortem-stage validation is not avail-

Placeholder for Figure 5. Subject-level confusion and error profile.

Panels to draw: **(a)** AIBL heldout subject-level confusion matrix; **(b)** true-to-predicted alluvial flow; **(c)** error subtype bars; **(d)** decision confidence by outcome; **(e)** decision margin by outcome; **(f)** clinical severity profiles; **(g)** atlas structural profiles for correct and boundary-error groups.

Figure 5: Subject-level error structure and disease-boundary behavior. **(a)** The AIBL held-out subject-level confusion matrix contains 216 endpoint units: CN 148/154 correctly classified, MCI 24/35 correctly classified, and AD 23/27 correctly classified. **(b)** The true-to-predicted flow shows that most CN endpoints remain CN, MCI errors split mainly toward AD rather than CN, and AD errors move to MCI rather than CN. **(c)** Error subtype bars quantify MCI→CN errors (2/35, 0.057), MCI→AD errors (9/35, 0.257), AD→MCI errors (4/27, 0.148), and AD→CN errors (0/27, 0.000). **(d)** Confidence distributions compare correctly classified endpoints against boundary-error endpoints using the maximum subject-level predicted probability. **(e)** Decision-margin distributions evaluate whether errors occur near small probability margins between adjacent disease classes. **(f)** Clinical severity profiles compare MMSE, CDR-SB, age, and APOE4 across correct MCI, MCI→AD, correct AD, and AD→MCI groups, testing whether boundary errors follow disease-severity gradients. **(g)** Atlas structural profiles compare hippocampal volume, lateral ventricular volume, and AD-like atlas z-score across the same groups. Residual mistakes concentrate near CN/MCI/AD disease boundaries rather than representing a collapse of AD cases into CN.

Placeholder for Figure 6. Algorithmic ablation, calibration, risk tradeoff, and stability.

Panels to draw: **(a)** ablation BAcc bars; **(b)** ablation MCI and AD recall; **(c)** AD-to-CN error comparison; **(d)** calibration ECE/NLL or reliability curve; **(e)** risk tradeoff curve; **(f)** leave-one-model-out stability; **(g)** bootstrap uncertainty forest plot.

Figure 6: Algorithmic ablation, calibration, risk tradeoff, and stability of RC-SPE. **(a)** AIBL BAcc is shown for best single base model (0.756), arithmetic mean ensemble (0.711), equal log-pooling (0.648), final weights only (0.815), weights+offsets (0.823), weights+temperature (0.815), full RC-SPE scan-level (0.810), and full RC-SPE subject-level (0.833). **(b)** MCI and AD recall are shown for the same variants, emphasizing that full subject-level RC-SPE balances MCI recall 0.686 and AD recall 0.852 better than simple averaging or equal log-pooling. **(c)** AD-to-CN error counts show risk reduction from 2 errors for the best single base model, 3 for arithmetic mean, and 5 for equal log-pooling to 0 for full RC-SPE. **(d)** Calibration compares expected calibration error and negative log-likelihood, including best single ECE 0.122/NLL 0.416 and full RC-SPE ECE 0.078/NLL 0.320; a reliability curve can be included when space allows. **(e)** The risk tradeoff panel compares the locked RC-SPE profile against MCI-rescue and AD-rescue alternatives, showing that the MCI-rescue profile reaches MCI recall 0.886 but reduces AD recall to 0.593 and IXI CN retention to 0.959. **(f)** Leave-one-model-out analysis shows that dropping any one of the six base streams preserves AIBL BAcc between 0.823 and 0.835 and keeps AD-to-CN errors at 0. **(g)** Bootstrap uncertainty reports AIBL subject-level BAcc 0.833 with 95% CI 0.759–0.899, MCI recall 0.686 with CI 0.531–0.839, and AD recall 0.852 with CI 0.710–0.966. Together, these panels establish RC-SPE as a risk-constrained algorithmic contribution rather than a generic model average.

able.

OASIS remained weak without OASIS tuning. The final subject-level model achieved OASIS accuracy 0.586, BAcc 0.334, macro AUC 0.554, AD-vs-CN AUC 0.371, and CN/MCI/AD recall 0.966/0.034/0.000. This result is reported as an unresolved transfer limitation, not as successful external validation.

4 Discussion

4.1 Subject-level external AD staging as the central claim

This study is best interpreted as a domain-adapted, subject-level external AD staging framework. The central evidence is not a single scan-level accuracy number, but the combination of locked AIBL heldout CN/MCI/AD classification, IXI healthy-control specificity, bootstrap uncertainty, error analysis, algorithmic ablation, and atlas-based structural neurodegeneration consistency.

The design makes the endpoint hierarchy explicit. AIBL heldout is the primary external staging endpoint, IXI tests whether healthy subjects are overcalled as impaired, and OASIS is retained as an external stress test. This structure gives each dataset a defined inferential role and reduces ambiguity when external datasets behave differently.

4.2 Interpretation of the final model

The final model is strongest when interpreted as a domain-adapted external AD staging framework. AIBL adaptation data were used for model fitting and calibration, but AIBL heldout endpoint units remained locked and were not used for final endpoint reporting. This distinction is important. The work does not solve pure ADNI-to-AIBL zero-shot staging, but it does demonstrate that anatomically grounded MRI features and core clinical variables can support robust heldout subject-level staging within an external cohort.

The algorithmic contribution is RC-SPE rather than a new attention block. Its novelty is methodological and risk-oriented: it converts heterogeneous base-model probability streams into a locked subject-level endpoint while explicitly balancing external BAcc, MCI/AD recall, AD-to-CN error avoidance, probability calibration, and IXI healthy specificity. The ablation results show that simple averaging or equal log-pooling are insufficient, while the full subject-level RC-SPE improves AIBL BAcc and removes AD-to-CN errors. The risk-profile experiment also shows why the highest MCI-recall candidate was not selected: it increased healthy-control false impairment in IXI.

RC-SPE is therefore a pragmatic solution to a clinically important modeling problem. Single models can perform well on aggregate metrics while failing minority disease classes or producing false impairment in healthy controls. The locked algorithm explicitly balances ex-

Placeholder for Figure 7. Atlas-based structural neurodegeneration consistency.

Panels to draw: **(a)** AD-key region definition; **(b)** region-level structural-effect ranking; **(c)** AIBL heldout consistency test; **(d)** all-labeled consistency test; **(e)** cohort comparison across AIBL heldout, all-labeled, and ADNI-only; **(f)** structural pattern map; **(g)** interpretation boundary for atlas structural neurodegeneration consistency.

Figure 7: **Atlas-based structural neurodegeneration consistency.** **(a)** The a priori AD-key atlas set identifies bilateral hippocampus, bilateral amygdala, and bilateral lateral ventricles as the structural regions used for the consistency test. **(b)** Region-level structural-effect ranking displays all atlas regions by AD-related volume or structural difference, allowing the AD-key set to be compared with the full regional background. **(c)** In the locked AIBL heldout endpoint, the AD-key atlas-volume consistency score is 0.510 compared with a uniform regional null of 0.286, with bootstrap CI 0.479–0.526 and permutation $p = 0.026$. **(d)** Across all labeled AD-relevant data, the AD-key consistency score is 0.426 versus the same null of 0.286, with permutation $p = 0.0207$. **(e)** A cohort-comparison panel shows AIBL heldout, all-labeled, and ADNI-only checks side by side; the ADNI-only internal check remains non-significant with score 0.342 and $p = 0.1843$. **(f)** The structural pattern map emphasizes hippocampal/amygdala atrophy and lateral ventricular enlargement as the intended MRI neurodegeneration pattern. **(g)** The interpretation boundary states that this is atlas structural neurodegeneration consistency, not direct Braak-stage validation, not an attention-map biomarker claim, and not clinical-device evidence.

ternal BAcc, MCI/AD recall, and IXI healthy specificity. Its value is therefore not only the final BAcc number, but also the combination of external heldout performance, zero IXI false impairment, bootstrap uncertainty, calibration behavior, and interpretable error behavior.

4.3 MCI/AD boundary behavior

The error analysis clarifies the clinical meaning of the remaining mistakes. On AIBL heldout, AD endpoint units were not missed as CN; residual AD errors were classified as MCI. MCI errors were split between correct MCI and AD, with only two MCI endpoint units classified as CN. This pattern is preferable to a model that preserves overall accuracy by collapsing minority disease classes into CN, but it also shows that precise MCI/AD boundary staging remains challenging.

MCI is biologically and clinically heterogeneous. Some MCI participants may show AD-like neurodegeneration, while others may remain stable or have non-AD etiologies. A subject-level classifier operating on retrospective structural MRI and clinical variables is therefore expected to show uncertainty near the MCI/AD boundary. The manuscript makes this limitation explicit rather than hiding it behind aggregate accuracy.

4.4 Biological consistency without overclaiming

The biological analysis is intentionally bounded. The AIBL heldout atlas-volume consistency result supports disease-consistent structural MRI change in AD-relevant regions. It does not demonstrate explanation-map biomarkers, and it does not provide postmortem-stage validation. This narrower claim is more defensible and better aligned with the available data.

The atlas-based biological analysis is a structural MRI consistency test. It evaluates whether AD-related atlas volume changes concentrate in a priori AD-relevant regions compared with a transparent regional null. Without neuropathology labels, the result is not interpreted as direct pathological-stage validation.

4.5 OASIS and internal calibration limitations

Several limitations remain. OASIS transfer is not solved and is treated as a stress-test limitation requiring larger and cleaner external cohorts. The internal subject-level calibration pattern remains modest, especially for CN specificity within the internal split. Clinical variables contain substantial diagnostic signal, as shown by the clinical-only comparator. The 21-region atlas is anatomically interpretable but coarse, and finer parcellations may better capture cortical AD patterns.

These limitations do not invalidate the locked AIBL heldout result, but they define its correct scope. The model supports domain-adapted external heldout staging under the revised

protocol. It does not support universal generalization across all public AD datasets, and it should not be described as deployment-ready clinical software.

4.6 Open science and clinical translation boundary

The revised repository is designed for reproducible research use. It includes code for cohort construction, feature handling, model evaluation, probability ensembling, bootstrap analysis, figure generation, documentation, and a lightweight research API/frontend. This improves transparency and helps readers audit the claims. However, raw neuroimaging and clinical data remain subject to original data-use agreements and cannot be redistributed directly.

Clinically, the model is best viewed as a decision-support research prototype. Prospective multi-center validation, scanner/protocol robustness testing, local calibration, integration with clinical workflows, uncertainty reporting, and regulatory review would be required before any clinical use. The manuscript therefore separates deployability as software from clinical deployability as a medical product.

5 Conclusion

This work presents an atlas-guided multimodal AD staging framework centered on RC-SPE, locked AIBL heldout subject-level validation, IXI healthy-control specificity, algorithmic ablation, bootstrap uncertainty, MCI/AD error analysis, and atlas-based structural neurodegeneration consistency. The strongest supported claim is domain-adapted external AD staging with an MRI neurodegeneration proxy, not pure zero-shot generalization, postmortem-stage validation, OASIS success, or clinical deployment readiness.

Data Availability

Raw ADNI, AIBL, OASIS, and IXI data are governed by their original data-use agreements and are not redistributed. Public repository files are limited to code, aggregate reports, figures, documentation, final model configuration, and toy probability examples. Row-level subject/scan predictions, private clinical spreadsheets, MRI volumes, and model checkpoints are excluded from the public package.

Code Availability

The public research implementation, documentation, aggregate reports, deployment wrappers, and final ensemble configuration are available at <https://github.com/Lava168/ARA-Net>. The released software is a research prototype; it is not a medical device and is not cleared or approved for clinical use.

Author Contributions (CRediT)

Yuanqin Zhao: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Writing – Original Draft, Visualization. **Yu Zhang:** Methodology, Software, Validation, Writing – Review & Editing. **Jinghong Hu:** Investigation, Data Curation, Writing – Review & Editing. **Jian Mao:** Methodology, Validation, Writing – Review & Editing. **Bisong Zhou:** Data Curation, Writing – Review & Editing. **Zhuoyao Song:** Data Curation, Writing – Review & Editing. **Tengfei Guo:** Conceptualization, Supervision, Funding Acquisition, Writing – Review & Editing. **Shaohua Ma:** Conceptualization, Supervision, Funding Acquisition, Project Administration, Writing – Review & Editing.

Acknowledgments

Data collection and sharing for this project was funded by the Alzheimer’s Disease Neuroimaging Initiative (ADNI; National Institutes of Health Grant U01 AG024904) and DOD ADNI (Department of Defense award number W81XWH-12-2-0012). ADNI is funded by the National Institute on Aging, the National Institute of Biomedical Imaging and Bioengineering, and through generous contributions from multiple agencies and private partners (a complete listing is available at <https://adni.loni.usc.edu>). The IXI dataset was collected as part of the IXI – Information eXtraction from Images project funded by EPSRC GR/S21533/02. The OASIS project was supported by grants P50 AG05681, P01 AG03991, P01 AG026276, R01 AG021910, and UL1 TR000448 from the National Institutes of Health.

Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This work was supported by the Shenzhen Bay Laboratory [grant number to be inserted], the National Natural Science Foundation of China [grant number to be inserted], and the Tsinghua Shenzhen International Graduate School.

References

World Health Organization, Dementia fact sheet, 2023. <https://www.who.int/news-room/fact-sheets/detail/dementia>

440 C.R. Jack, D.A. Bennett, K. Blennow, M.C. Carrillo, B. Dunn, S.B. Haeberlein, et al., NIA-AA
 441 research framework: toward a biological definition of Alzheimer’s disease, *Alzheimer’s &*
 442 *Dementia* 14 (4) (2018) 535–562. <https://doi.org/10.1016/j.jalz.2018.02.018>

443 S. Korolev, A. Safiullin, M. Belyaev, Y. Dodonova, Residual and plain convolutional neural
 444 networks for 3D input volumes, arXiv preprint arXiv:1701.06643, 2017.

445 A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, et al., An
 446 image is worth 16x16 words: transformers for image recognition at scale, in: *ICLR*, 2020.

447 H. Jia, et al., STNet: spatial-temporal network for Alzheimer’s disease diagnosis, in: *Proceed-*
 448 *ings of MICCAI*, Springer, 2023.

449 J. Gao, et al., Robust LSTM-based classification for Alzheimer’s disease, in: *Proceedings of*
 450 *IEEE EMBC*, 2023.

451 M. Alkhathami, et al., ECAResNet with SMOTE and focal loss for Alzheimer’s disease classi-
 452 fication, *Biomedical Signal Processing and Control* 99 (2025) 106882. [https://doi.org/](https://doi.org/10.1016/j.bspc.2024.106882)
 453 [10.1016/j.bspc.2024.106882](https://doi.org/10.1016/j.bspc.2024.106882)

454 S. Murugan, et al., DEMNET: a deep learning model for early diagnosis of Alzheimer’s dis-
 455 eases and dementia from MRI images, *Biomedical Signal Processing and Control* 65 (2021)
 456 102386. <https://doi.org/10.1016/j.bspc.2020.102386>

457 A. Majee, et al., 3D HCCT: a hybrid CNN-convolutional transformer for brain MRI classifi-
 458 cation, *Pattern Recognition* 148 (2024) 110197. [https://doi.org/10.1016/j.patcog.](https://doi.org/10.1016/j.patcog.2023.110197)
 459 [2023.110197](https://doi.org/10.1016/j.patcog.2023.110197)

460 X. Zhang, et al., THAN: task-driven hierarchical attention network for the diagnosis of
 461 Alzheimer’s disease, *NeuroImage* 253 (2022) 119099. [https://doi.org/10.1016/j.](https://doi.org/10.1016/j.neuroimage.2022.119099)
 462 [neuroimage.2022.119099](https://doi.org/10.1016/j.neuroimage.2022.119099)

463 R.R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, D. Batra, Grad-CAM: visual
 464 explanations from deep networks via gradient-based localization, in: *Proceedings of ICCV*,
 465 IEEE, 2017, pp. 618–626.

466 S.M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, in: *Advances*
 467 *in Neural Information Processing Systems (NeurIPS)*, 2017.

468 M. Sundararajan, A. Taly, Q. Yan, Axiomatic attribution for deep networks, in: *Proceedings of*
 469 *ICML*, 2017.

470 J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, B. Kim, Sanity checks for saliency
 471 maps, in: *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

- C. Rudin, Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead, *Nature Machine Intelligence* 1 (5) (2019) 206–215. <https://doi.org/10.1038/s42256-019-0048-x>
- J. Amann, A. Blasimme, E. Vayena, D. Frey, V.I. Madai, Explainability for artificial intelligence in healthcare: a multidisciplinary perspective, *BMC Medical Informatics and Decision Making* 20 (2020) 310. <https://doi.org/10.1186/s12911-020-01332-6>
- G. Varoquaux, P.R. Raamana, D.A. Engemann, A. Hoyos-Idrobo, Y. Schwartz, B. Thirion, Assessing and tuning brain decoders: cross-validation, caveats, and guidelines, *NeuroImage* 145 (2017) 166–179. <https://doi.org/10.1016/j.neuroimage.2016.10.038>
- H. Braak, E. Braak, Neuropathological staging of Alzheimer-related changes, *Acta Neuropathologica* 82 (4) (1991) 239–259. <https://doi.org/10.1007/BF00308809>
- L. Henschel, S. Conjeti, S. Estrada, K. Diers, B. Fischl, M. Reuter, FastSurfer—a fast and accurate deep learning based neuroimaging pipeline, *NeuroImage* 219 (2020) 117012. <https://doi.org/10.1016/j.neuroimage.2020.117012>
- Z. Zhou, V. Sodha, J. Pang, M.B. Gotway, J. Liang, Models Genesis: generic autodidactic models for 3D medical image analysis, *Medical Image Analysis* 67 (2021) 101840. <https://doi.org/10.1016/j.media.2020.101840>
- I. Loshchilov, F. Hutter, Decoupled weight decay regularization, in: *Proceedings of ICLR*, 2019.
- H. Zhang, M. Cisse, Y.N. Dauphin, D. Lopez-Paz, mixup: beyond empirical risk minimization, in: *Proceedings of ICLR*, 2018.
- European Commission, Proposal for a regulation laying down harmonised rules on artificial intelligence (AI Act), COM/2021/206, 2021.
- S.P. Poulin, R. Dautoff, J.C. Morris, L.F. Barrett, B.C. Dickerson, Amygdala atrophy is prominent in early Alzheimer’s disease and relates to symptom severity, *Psychiatry Research: Neuroimaging* 194 (1) (2011) 7–13. <https://doi.org/10.1016/j.psychresns.2011.06.014>
- G. Livingston, J. Huntley, A. Sommerlad, D. Ames, C. Ballard, S. Banerjee, et al., Dementia prevention, intervention, and care: 2020 report of the Lancet Commission, *The Lancet* 396 (10248) (2020) 413–446. [https://doi.org/10.1016/S0140-6736\(20\)30367-6](https://doi.org/10.1016/S0140-6736(20)30367-6)
- B. Dubois, H. Hampel, H.H. Feldman, P. Scheltens, P. Aisen, S. Andrieu, et al., Preclinical Alzheimer’s disease: definition, natural history, and diagnostic criteria, *Alzheimer’s & Dementia* 12 (3) (2016) 292–323. <https://doi.org/10.1016/j.jalz.2016.02.002>

- R.C. Petersen, O. Lopez, M.J. Armstrong, T.S.D. Getchius, M. Ganguli, D. Gloss, et al., Practice guideline update summary: mild cognitive impairment, *Neurology* 90(3) (2018) 126–135. <https://doi.org/10.1212/WNL.0000000000004826>
- G.B. Frisoni, N.C. Fox, C.R. Jack, P. Scheltens, P.M. Thompson, The clinical use of structural MRI in Alzheimer disease, *Nature Reviews Neurology* 6(2) (2010) 67–77. <https://doi.org/10.1038/nrneurol.2009.215>
- J. Wen, E. Thibeau-Sutre, M. Diaz-Melo, J. Samper-González, A. Routier, S. Bottani, et al., Convolutional neural networks for classification of Alzheimer’s disease: overview and reproducible evaluation, *Medical Image Analysis* 63 (2020) 101694. <https://doi.org/10.1016/j.media.2020.101694>
- T. Jo, K. Nho, A.J. Saykin, Deep learning in Alzheimer’s disease: diagnostic classification and prognostic prediction using neuroimaging data, *Frontiers in Aging Neuroscience* 11 (2019) 220. <https://doi.org/10.3389/fnagi.2019.00220>
- G. Huang, Z. Liu, L. Van Der Maaten, K.Q. Weinberger, Densely connected convolutional networks, in: *Proceedings of CVPR, IEEE*, 2017, pp. 2261–2269.
- Z.C. Lipton, The mythos of model interpretability, *Queue* 16(3) (2018) 31–57. <https://doi.org/10.1145/3236386.3241340>
- R. Guidotti, A. Monreale, S. Ruggieri, F. Turini, F. Giannotti, D. Pedreschi, A survey of methods for explaining black box models, *ACM Computing Surveys* 51(5) (2019) 93:1–93:42. <https://doi.org/10.1145/3236009>
- A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, et al., Attention is all you need, in: *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 5998–6008.
- D. Bahdanau, K. Cho, Y. Bengio, Neural machine translation by jointly learning to align and translate, in: *Proceedings of ICLR*, 2015.
- B. Fischl, FreeSurfer, *NeuroImage* 62(2) (2012) 774–781. <https://doi.org/10.1016/j.neuroimage.2012.01.021>
- N. Tzourio-Mazoyer, B. Landeau, D. Papathanassiou, F. Crivello, O. Etard, N. Delcroix, et al., Automated anatomical labeling of activations in SPM using a macroscopic anatomical parcellation of the MNI MRI single-subject brain, *NeuroImage* 15(1) (2002) 273–289. <https://doi.org/10.1006/nimg.2001.0978>
- R.S. Desikan, F. Ségonne, B. Fischl, B.T. Quinn, B.C. Dickerson, D. Blacker, et al., An automated labeling system for subdividing the human cerebral cortex on MRI scans into gyral

based regions of interest, *NeuroImage* 31 (3) (2006) 968–980. <https://doi.org/10.1016/j.neuroimage.2006.01.021>

S. Parisot, S.I. Ktena, E. Ferrante, M. Lee, R. Guerrero, B. Glocker, D. Rueckert, Disease prediction using graph convolutional networks: application to autism spectrum disorder and Alzheimer’s disease, *Medical Image Analysis* 48 (2018) 117–130. <https://doi.org/10.1016/j.media.2018.06.001>

X. Li, Y. Zhou, N. Dvornek, M. Zhang, S. Gao, J. Zhuang, et al., BrainGNN: interpretable brain graph neural network for fMRI analysis, *Medical Image Analysis* 74 (2021) 102233. <https://doi.org/10.1016/j.media.2021.102233>

C.R. Jack, M.A. Bernstein, N.C. Fox, P. Thompson, G. Alexander, D. Harvey, et al., The Alzheimer’s Disease Neuroimaging Initiative (ADNI): MRI methods, *Journal of Magnetic Resonance Imaging* 27 (4) (2008) 685–691. <https://doi.org/10.1002/jmri.21049>

D.S. Marcus, T.H. Wang, J. Parker, J.G. Csernansky, J.C. Morris, R.L. Buckner, Open Access Series of Imaging Studies (OASIS): cross-sectional MRI data in young, middle aged, nondemented, and demented older adults, *Journal of Cognitive Neuroscience* 19 (9) (2007) 1498–1507. <https://doi.org/10.1162/jocn.2007.19.9.1498>

J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed., Lawrence Erlbaum Associates, Hillsdale, NJ, 1988.

P. Scheltens, K. Blennow, M.M.B. Breteler, B. de Strooper, G.B. Frisoni, S. Salloway, W.M. Van der Flier, Alzheimer’s disease, *The Lancet* 388 (10043) (2016) 505–517. [https://doi.org/10.1016/S0140-6736\(15\)01124-1](https://doi.org/10.1016/S0140-6736(15)01124-1)

B.C. Dickerson, I. Goncharova, M.P. Sullivan, C. Forchetti, R.S. Wilson, D.A. Bennett, et al., MRI-derived entorhinal and hippocampal atrophy in incipient and very mild Alzheimer’s disease, *Neurobiology of Aging* 22 (5) (2001) 747–754. [https://doi.org/10.1016/S0197-4580\(01\)00271-8](https://doi.org/10.1016/S0197-4580(01)00271-8)

A. Delacourte, J.P. David, N. Sergeant, L. Buée, A. Wattez, P. Vermersch, et al., The biochemical pathway of neurofibrillary degeneration in aging and Alzheimer’s disease, *Neurology* 52 (6) (1999) 1158–1165. <https://doi.org/10.1212/WNL.52.6.1158>

J.-P. Fortin, D. Parker, B. Tunc, T. Watanabe, M.A. Elliott, K. Ruparel, et al., Harmonization of multi-site diffusion tensor imaging data, *NeuroImage* 161 (2017) 149–170. <https://doi.org/10.1016/j.neuroimage.2017.08.047>

R. Pomponio, G. Erus, M. Habes, H. Doshi, D. Srinivasan, E. Mamourian, et al., Harmonization of large MRI datasets for the analysis of brain imaging patterns throughout the lifespan, *NeuroImage* 208 (2020) 116450. <https://doi.org/10.1016/j.neuroimage.2019.116450>

- 571 E. Moradi, A. Pepe, C. Gaser, H. Huttunen, J. Tohka, Machine learning framework for early
572 MRI-based Alzheimer's conversion prediction in MCI subjects, *NeuroImage* 104 (2015)
573 398–412. <https://doi.org/10.1016/j.neuroimage.2014.10.002>
- 574 C.R. Jack, D.A. Bennett, K. Blennow, M.C. Carrillo, H.H. Feldman, G.B. Frisoni, et al., A/T/N:
575 an unbiased descriptive classification scheme for Alzheimer disease biomarkers, *Neurology*
576 91 (3) (2018) 111–116. <https://doi.org/10.1212/WNL.0000000000005560>
- 577 E.H. Corder, A.M. Saunders, W.J. Strittmatter, D.E. Schmechel, P.C. Gaskell, G.W. Small, et al.,
578 Gene dose of apolipoprotein E type 4 allele and the risk of Alzheimer's disease in late on-
579 set families, *Science* 261 (5123) (1993) 921–923. [https://doi.org/10.1126/science.](https://doi.org/10.1126/science.8346443)
580 [8346443](https://doi.org/10.1126/science.8346443)
- 581 A. Schaefer, R. Kong, E.M. Gordon, T.O. Laumann, X.-N. Zuo, A.J. Holmes, et al., Local-
582 global parcellation of the human cerebral cortex from intrinsic functional connectivity MRI,
583 *Cerebral Cortex* 28 (9) (2018) 3095–3114. <https://doi.org/10.1093/cercor/bhx179>